Department of Economics

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Correlated Shocks: A New Perspective on Financial Repression

Adriano Mariani

May 30, 2023

Abstract

In this paper, I investigate optimal monetary policy strategies in response to a fiscal shock that raises the real debt stock under two separate policy frameworks; optimal Ramsey policy and an interest rate smoothing Taylor rule. I find that the policy response in the baseline model is very different under the two frameworks. In the former case, policy works through the ex-ante channel (financial repression), while in the latter case, policy acts primarily through the ex-post channel. In part II, I extend the New Keynesian model by Sheedy, 2022 by including a demand/preference shock to more accurately reflect the dynamics of COVID-19 in advanced economies. I study optimal monetary policy in response to a correlated realization of the two shocks. I find that, under optimal Ramsey policy, the policy response in the two-shock model is approximately 3 times as large in magnitude relative to the baseline model. However, under interest rate smoothing, the policy response across the two models is very different, requiring policy rate hikes in the baseline case vs. policy rate cuts in the two-shock model.

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1 Introduction

The COVID-19 crisis has at large been described as a once-in-a-century event, with its effect on government finances often likened in magnitude to major wars. The virus, and especially the measures put in place to contain its spread, caused tremendous economic hardship, with the unemployment rate in the US touching a post-war record high of 16%. In response to this event, fiscal authorities in advanced economies (AE's) embarked on unprecedented fiscal stimulus in both scope and scale. For instance, in the US, the federal government, amongst other measures, approved several rounds of direct

payments to households and expanded unemployment benefits [mis20], to name a few. To finance said expenditures at a time of reduced tax receipts, governments issued debt at the highest rate in the post-war period. This makes the topic of financial repression, which is defined as "policies resulting in savers earning returns below the rate of inflation", ie reducing the ex-ante real return, highly topical and of significant practical interest.

In this regard, this paper analyzes optimal monetary policy in response to an exogenous fiscal shock which raises the real value of the government's debt outstanding. The size of the shock is assumed to be too large in magnitude to be absorbed by raising taxes, which forces the government to tap the debt markets in order to finance the higher expenditures.

The paper is structured as follows. Part I lays out the baseline model by Sheedy, 2022, which I extend by accounting for potential persistence of the fiscal shock. I calibrate the model and explore the dynamics of the optimal monetary policy (Ramsey policy) response to the fiscal shock. I proceed to do a sensitivity analysis around the main parameters, in order to assess whether the results are robust to my calibration. These are δ, ρ_v, σ, θ and ψ. I define a 'main parameter' as those which are of significant importance for either the direction or the magnitude of the optimal policy response to the shock. In part II, I augment the model by including a demand/preference shock. The idea is motivated by the fact that in practice, the fiscal shock observed in the wake of the COVID-19 crisis was in response to the demand shock caused by the crisis in the first place. Hence not including a contemporaneous negative demand shock would lead to misleading inferences surrounding the optimal policy response to the fiscal shock. In this vein, I analyze optimal monetary policy (Ramsey policy) in the face of a correlated realization of the two shocks. My calibration strategy for the parameters not present in the baseline model, namely Ψ and ρ_d is detailed in Section 3.2. All the other parameters are the same across the two models to facilitate comparisons. Finally, I perform a sensitivity analysis for ρ_d, the only parameter not present in the baseline model.

In both parts I and II, I assess optimal policy in response to the shock under two different lenses. First, following the literature on optimal monetary policy with commitment [Woo04], I assume the central bank commits at the time of the shock to a path for the nominal interest rate which minimizes the welfare loss function described in Section 2.3.1. This approach however has a major drawback. Under a quadratic welfare loss function, large but short-lived deviations from steady-state levels are preferred to persistently small deviations. Given there is no discounting of future welfare losses, welfare losses very far into the future bear the same weight as welfare losses today in policymakers' minds. This leads to large (often infeasibly large in practice) and abrupt changes in policy instruments, which is at odds with the principle of graduality and interest rate smoothing observed in the setting of monetary policy in practice [Woo03]. The literature has proposed a variety of explanations for policymakers' desire for graduality. These include financial stability considerations [Goo87], measurement errors in key macro variables, and uncertainty in relevant structural parameters [SW00], to cite a few. Moreover, [Woo03] finds that in the absence of full commitment, introducing a preference for low volatility in policy rates leads to time-consistent policies which achieve outcomes close to those achievable through full commitment. This finding motivates our choice for the formulation of an alternative 'optimal policy rule', the classic specification of the policy-smoothing Taylor rule as in [HB12], which is set out in Section 2.3.2.

2 Part I: Optimal Policy in the Baseline model

2.1 (i) Linearized Equilibrium

The Baseline model follows Sheedy, 2022. In addition, I account for potential persistence in the fiscal shock by assuming it follows an AR(1) process (equation 7). The linearized model is then:

$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t \tag{NKPC}$$

$$y_t = E_t y_{t+1} - \sigma r_t \tag{IS}$$

$$r_t = E_t R_{t+1} \tag{3}$$

$$j_t = \beta\delta E_t j_{t+1} + (1-\beta\delta) i_t \tag{4}$$

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{1-\beta\delta}(j_t - j_{t-1}) \tag{5}$$

$$\beta d_t = d_{t-1} + R_t + \beta v_t \tag{6}$$

$$v_t = \rho_v v_{t-1} + \phi_t \tag{7}$$

Where:

$$\kappa = \upsilon\gamma\xi$$

$$\upsilon = \frac{1}{1+\epsilon\eta}$$

$$\gamma = \frac{1}{\sigma} + \frac{1+\eta}{\psi} + \eta$$

$$\xi = \frac{(1-\theta)(1-\theta\beta)}{\theta}$$

$$\eta = \frac{\alpha}{(1-\alpha)}$$

There are 8 endogenous variables: inflation π_t, the output gap y_t, the ex-ante real interest rate r_t, the bond yield j_t, the ex-post real return on bonds R_t, the real value of government debt d_t, the nominal interest rate i_t and finally the AR(1) process governing the persistence of the fiscal shock v_t. The fiscal shock itself ϕ_t is exogenous. We have 8 endogenous variables in 7 equations, ie one degree of freedom. We can close the model by specifying a monetary policy rule.

2.2 (ii) Calibration

I now move on to calibrate the model. I will calibrate the model to the US economy for three main reasons. Firstly, due to its current and past status as the world's largest economy, there is an extensive body of literature calibrating DSGE models to the US economy, see for instance [SW07]. Secondly, and partly as a result of the first point, it is the economy with the highest quality and widest range of economic data available to the public. Thirdly, it makes sense to frame the policy question around the US economy as it experienced perhaps the largest fiscal shock amongst advanced economies (AE's) [mis20]. The total size of the fiscal response in the first year following the COVID shock was 5.2 trillion US dollars [Rom21], approximately 25% of pre-COVID GDP. The following table summarizes the parameter values of the baseline calibration, with a bulleted explanation of the intuition behind the chosen values following.

Parameter	Value
β	0.9984
σ	0.594
δ	0.981
ϵ	6
ψ	0.75
θ	$\frac{11}{12}$
ρ_v	0.6813
α	$\frac{1}{3}$
ρ_i	0.7
Φ_π	0.5
Φ_y	0.6

Table 1: Calibration for Baseline Model.

- I calibrate β to be consistent with a 2% p.a steady-state real interest rate $\bar{r} = \beta^{-1} - 1$. This translates into a monthly discount factor $\beta \approx 0.9984$.
- I follow [HHIR15] and set $\sigma = 0.594$, which is the mean estimate of the IES for the US economy across 1429 studies. Given there is substantial uncertainty in the literature surrounding the estimates of this parameter and its key role in the transmission of monetary policy, I will study the cases of a lower and higher value of σ relative to the above baseline.

- δ is calibrated to match the weighted average Macaulay duration of the total stock of US government debt, which I find is ≈ 4 years (48 months) using data from the federal government's database. In our model, the duration of the debt stock is given by $\frac{1}{1-\beta\delta}$. Thus, given our parametrization for β, we get $\delta \approx 0.981$. I will also experiment with the limiting cases where the long-term bond is, respectively, a perpetuity: $\delta = 1$, and a one-period bond $\delta = 0$.

- ϵ is the price elasticity of demand of monopolistically competitive firms and is calibrated to match a steady-state markup of 20% over marginal costs, as in [Ire04], which implies $\epsilon = 6$.

- ψ is the Frisch elasticity of Labor supply. Following [CGMW11], I set $\psi = 0.75$.

- I follow the findings from [NS08] and calibrate θ to match the average frequency of price adjustment, which is approximately once per year across all goods, ie $\theta = \frac{11}{12}$. In addition, I will experiment with the limiting cases of flexible prices: $\theta \approx 0$ and perfectly sticky prices: $\theta \approx 1$.

- ρ_v represents the persistence of the AR(1) process for the fiscal shock. ρ_v is the portion of the initial shock that carries over to the following period. Then the portion of the initial shock that carries over l periods after is $(\rho_d)^l$. Thus assuming that by the $t + l$th period the initial shock has faded completely up to some tolerance level (say 99% of the shock has faded), ρ_v then solves:

  $$(\rho_v)^l = 0.01$$

  To estimate l, the number of periods since the initial shock by which the shock has completely faded, I de-trend the total US government debt outstanding (data from FRED) using the HP filter as described in [KP+90]. I then plot deviations of the total debt outstanding from the HP-filtered trend. l is then the number of periods between the initial shock (Q2 2020) and when the debt outstanding returns to trend (Q2 2021), which implies $l = 12$. Using this estimate for l, we can use the bisection method to approximate the solution to the above non-linear equation and get $\rho_v \approx 0.6813$.

- I calibrate α, the implicit capital share in the aggregate production function to match the average post-war capital share of output in the US: $\alpha \approx \frac{1}{3}$. This implies that η, the elasticity of marginal cost with respect to firm-level output is $\eta = \frac{\alpha}{(1-\alpha)} = \frac{1}{2}$.

- For the parameters in the policy-smoothing Taylor rule of Sections 2.3.2 and 3.3.2, namely the weight placed by the central bank on i) inflation deviations from steady-state, ii) the output gap and iii) the persistence parameter respectively, I set $\Phi_\pi = 0.5$, $\Phi_y = 0.6$ and $\rho_i = 0.7$, which are the mean parameter estimates across advanced economies found by [HB12].

2.3 (iii) Simulation: Optimal Monetary Policy

2.3.1 (a) Optimal Ramsey Policy

In determining optimal monetary policy, the central bank chooses a path for the nominal interest rate $\{i_t\}_{t=0}^{\infty}$ by minimizing the utility-based welfare loss function as in Sheedy, 2022:

$$L_t = \frac{1}{2}\sum_{l=0}^{\infty}\beta^l E_t(\frac{\epsilon}{\upsilon\xi}\pi_{t+l}^2 + \gamma y_{t+l}^2)$$

Where υ, γ and ξ are defined in Section 2.1.

Figure 1 plots the impulse responses of the endogenous variables under optimal Ramsey policy to a fiscal shock equal to 10% of the steady-state debt stock. The optimal policy in response to the shock is to engineer financial repression by lowering policy rates aggressively on impact. The policy rate cuts feed through to economic activity and inflation via its impact on the ex-ante real rate, which affects the IS curve (equation 2) and the NKPC (equation 1). The combined effect of the policy cuts and the surprise inflation reduce the ex-ante real interest rate by $\approx 70\%$ annualized. The result is a $\approx 20\%$ annualized increase in inflation above steady-state levels and an $\approx 20\%$ rise in the output gap. The two variables then very sluggishly (especially inflation, which is still $\approx 4\%$ above steady-state levels at the end of the projection horizon) return to steady-state levels. Policy rates gradually

return to steady-state levels ≈ 2 years after the shock. Under the baseline calibration policymakers then temporarily raise policy rates slightly above steady-state levels to partially reign in the large rises in inflation and the output gap. The initial fall in policy rates feeds through to the bond yield which falls initially, causing the ex-post return on bonds to rise thanks to the capital gains for existing bondholders. Bond yields then rise above steady-state levels on expectations of policy firming, which coupled with the surprise inflation bites into bondholders' ex-post returns. The Bond yield does not converge to steady-state levels within the projection horizon (6 years). Interestingly, the shock raises the real value of the debt stock on impact by more than 1:1. This is mostly due to the persistence of the fiscal shock (as shown in Figure 5 in the appendix), but also due to the initial fall in bond yield, which raises the current value of the debt stock outstanding.

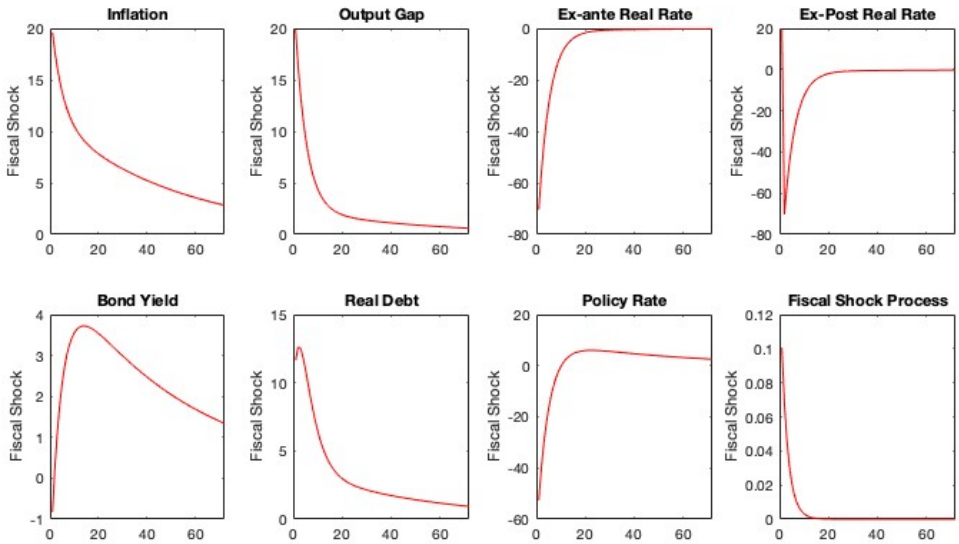

Figure 1: Optimal Ramsey Policy- Baseline Model

Thus the strategy behind the optimal policy is to "inflate away" the additional debt by engineering a steep fall in ex-ante real rates. Whilst it is definitely an interesting policy experiment, in practice this strategy may be infeasible for two main reasons.

- Firstly, a reduction in policy rates of the magnitude in question is likely to hit against the effective lower bound (ELB). Thus, in practice, the optimal monetary policy response requires some creativity on behalf of policymakers. An option would be to monetize the debt. To this end, an interesting extension to the current policy framework could be adding the lower bound constraint and allowing central banks to purchase a fraction of the newly-issued debt in the primary markets (monetization) or secondary markets (QE).

- Secondly, the policy response may be misleading as individuals may not be willing to hold the debt in the face of such a massive decrease in ex-ante real returns. In the model, the long-term bond is the only asset available to individuals to smooth consumption between periods, and hence they are in effect forced to hold it. To this end, another interesting extension could be adding a real asset not issued by the government, gold for instance, which would act as an imperfect substitute for the long-term bond. Even though in equilibrium the ex-ante expected return on both assets would have to be equalized, it may be optimal to substitute away from the long-term bond ex-post in the face of the steeply negative expected real returns.

2.3.2 (b) Policy-smoothed Taylor Rule

As argued in the introduction, in practice policymakers do not act in ways consistent with the minimization of a welfare loss function. Motivated by [Woo03]'s findings, I introduce the policy-smoothing

Taylor rule as an alternative 'optimal' policy rule, which takes into account the graduality principle. The rule takes the following linearized form:

$$i_t = \rho_i i_{t-1} + (1 - \rho_i)(\Phi_\pi \pi_t + \Phi_y y_t) \tag{8}$$

This would be equation 8 in the linearized equilibrium of Section 2.1. The calibration of the rule follows the principles set out in Section 2.2.

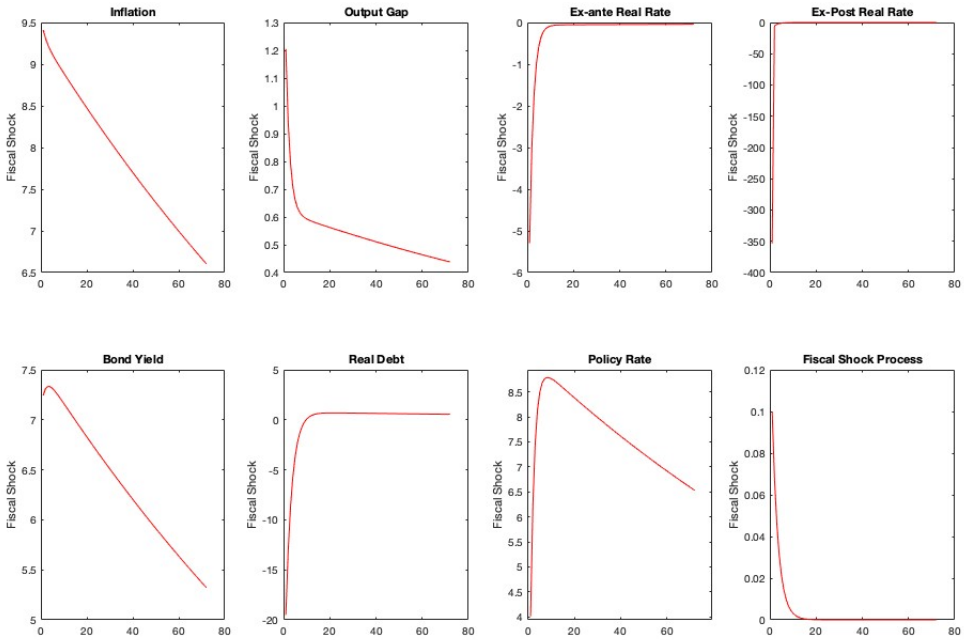

Figure 2: Policy smoothing Taylor Rule- Baseline Model

Figure 2 plots the impulse responses of the endogenous variables under this alternative policy formulation to a fiscal shock of the same size as in Section 2.3.1. Interestingly, under this policy framework, the real value of the debt stock falls in response to the fiscal shock. In contrast to the optimal Ramsey policy case, under interest rate smoothing policymakers raise interest rates on impact and in the months following the shock to $\approx 8.5\%$ above steady-state levels. This in turn raises bond yields and reduces dramatically the ex-post return on bonds. Essentially, policymakers act through the ex-post return channel rather than the ex-ante return channel: rather than cutting rates aggressively and reducing the real debt burden over time by engineering a reduction in ex-ante real rates, here policymakers raise rates aggressively to engineer a drop in the real value of debt on impact. This strategy is however at the cost of initially lower but more persistent inflation, which ends the projection horizon at $\approx 6\%$. This policy however leads to a more stable output gap, which only rises by 1.2% on impact, relative to the optimal Ramsey policy case.

The alternative policy formulation has two main advantages over optimal Ramsey policy:

- Firstly, under the policy-smoothed Taylor rule, policy rates are never set below steady-state levels in response to the shock, implying no risk of violating the ELB. This monetary policy strategy may thus be actually feasible in practice.
- Secondly, the rule is objectively easier to implement and does not require the central bank to commit to a set path for the policy rate ex-ante. This leads to time-consistent policies, as argued by [Woo03].

2.4 (iv) Sensitivity analysis

- Figure 7 in the appendix plots the IRFs in response to the fiscal shock under optimal Ramsey policy in the baseline model. Estimates of σ in the literature lie in the interval $(0, 10)$ [HHIR15]. I study 3 cases. First, in blue, I set $\sigma = 0.2$. The second, in red, is the baseline calibration of $\sigma = 0.594$. Lastly, in green, I set $\sigma_d = 1.5$. As $\sigma \to 0$, for a given value of the real rate households are less willing to intertemporally substitute consumption, thus a decrease of the ex-ante real rate of larger magnitude is required relative to the baseline calibration to absorb the fiscal shock. Thus low values of σ accentuate the optimal policy response relative to the baseline calibration. In the case of $\sigma = 1.5$, the optimal policy response is to raise policy rates marginally, in contrast to the first two cases. Interestingly, the policy response for large values of σ under optimal Ramsey policy is similar to the policy response under the interest rate smoothing Taylor rule of section 2.3.2 with the baseline calibration for σ. That is policy acts primarily through the ex-post return channel rather than through the ex-ante return channel.

- Figure 8 in the appendix plots the IRFs in response to the fiscal shock under optimal Ramsey policy in the baseline model. I study 3 cases, of which two are the limiting cases. First, in blue, I set $\delta = 0$, which corresponds to the limiting case of one-period bonds. The second, in red, is the baseline calibration of $\delta = 0.981$. Lastly, in green, I study the limiting case of a perpetuity and set $\delta = 1$. Intuitively, in the case of one-period bonds, the central bank is able to engineer financial repression by cutting policy rates aggressively in the period the shock hits the economy. This allows the government to finance the additional debt at steeply negative real rates. In the following period, the additional debt matures, which allows the central bank to bring policy rates back to steady-state levels immediately. In the perpetuity case, the additional debt never matures, hence it is intuitive that the optimal monetary policy strategy is acting through the ex-post return channel by raising policy rates and bond yields, which reduces the real value of the stock of debt on impact.

- Figure 9 in the appendix plots the IRFs in response to the fiscal shock under optimal Ramsey policy in the baseline model. I study 3 cases. First, in blue, I set $\psi = 0.4$, which is the mean micro estimate used by [Wha17]. The second, in red, is the baseline calibration of $\psi = 0.75$. Lastly, in green, I set $\psi = 2.84$, following [CGMW11], which is the average of several macro estimates of the Frisch elasticity of labor supply. Intuitively, low(high) values of ψ reduce(increase) the degree of real rigidities and hence the sensitivity of output to changes in the ex-ante real rates, thus a smaller(larger) decrease in policy rates relative to the baseline calibration is needed to reign in the shock. ψ thus only affects the magnitude of the optimal policy response, not its direction.

- Figure 10 in the appendix plots the IRFs in response to the fiscal shock under optimal Ramsey policy in the baseline model. I study 4 cases, of which 2 are the limiting cases of flexible prices and completely sticky prices. First, in blue, I set $\theta = 0.01$ to study the limiting case of flexible prices. Secondly, in light blue, I set $\theta = \frac{7}{8}$, which corresponds to a median price duration of 9 months, which is the lowest estimate found by [NS08]. The third, in red, is the baseline calibration of $\theta = \frac{11}{12}$. Lastly, in green, I study the limiting case of perfectly sticky prices and set $\theta = 0.99$. Under flexible prices, the optimal Ramsey policy works solely through the ex-post return channel, ie Inflation adjusts to keep the ex-ante real return fixed throughout the projection horizon, in stark contrast to the baseline calibration. As a result, and somewhat counterintuitively, under flexible prices inflation rises by less than in the baseline calibration but is more persistent. In the second case, the optimal policy works through a combination of the two channels. Finally, in case 4, when prices are very sticky, optimal policy works through the ex-ante return channel. The decrease in policy rates on impact is somewhat lower, but much more persistent as policy rates stand at $\approx 10\%$ below steady-state values by the end of the projection horizon. The calibration of θ thus affects both the direction and the magnitude of the optimal policy response.

- Figure 11 in the appendix plots the IRFs in response to the fiscal shock under optimal Ramsey policy in the baseline model. I study 3 cases. First, in blue, I set $\rho_v = 0$ to study the case with no persistence in the fiscal shock, ie the model from Sheedy, 2022. The second, in red, is the baseline calibration of $\rho_v = 0.6813$. Lastly, in green, I set $\rho_v = 0.8254$, which corresponds to a value of $l = 24$, ie the persistence consistent with the fiscal shock taking double the time relative to the baseline calibration to (almost) fade away completely. In the case of no persistence, policy

rates need to fall by a lower amount to reign in the shock, which is intuitive. As $\rho_v \to 1$, an ever stronger response of policy rates is needed to reign in the shock. This leads to a larger rise in inflation and the output gap, which requires policy rates to be lifted more significantly above steady-state levels approximately 10 periods after the initial shock. Thus, similarly to ψ, ρ_v affects only the magnitude, not the direction of the optimal policy response.

3 Part II: Optimal Policy in the Two-shock model

3.1 (i) Linearized Equilibrium

I now include a demand shock to the model which distorts the IS curve. The model is now:

$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t \tag{NKPC}$$

$$y_t = E_t y_{t+1} + \sigma r_t + d_t \tag{IS}$$

$$r_t = E_t R_{t+1} \tag{3}$$

$$j_t = \beta\delta E_t j_{t+1} + (1-\beta\delta) i_t \tag{4}$$

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{1-\beta\delta}(j_t - j_{t-1}) \tag{5}$$

$$\beta d_t = d_{t-1} + R_t + \beta v_t \tag{6}$$

$$v_t = \rho_v v_{t-1} + \phi_t \tag{7}$$

$$d_t = \rho_d d_{t-1} + \Psi \phi_t \tag{8}$$

We now have 9 endogenous variables, 8 from the baseline model in part I plus the AR(1) process governing the persistence of the demand shock d_t. Again, ϕ is the only exogenous variable. I still retain one degree of freedom to specify a monetary policy rule, which would close the model. In this setup, we can interpret the correlated realization of the shock ϕ_t as if it were a joint realization of two separate shocks hitting the economy contemporaneously. This modeling assumption is made for tractability, as it allows us to ignore the joint distribution of the two separate shocks over time, as highlighted in [CR20].

3.2 (ii) Calibration

In this version of the model, we have two additional parameters to calibrate relative to the baseline model in Part I. These are:

- I calibrate the persistence parameter of the AR(1) process governing the demand shock ρ_d in the same way as the persistence parameter for the AR(1) process governing the fiscal shock. ρ_d solves:

  $$(\rho_d)^l = 0.01$$

  I estimate l using the same method as in section 2.2, using deviations of Real GDP (chained 2012 US dollars) from its HP-filtered trend as described by [KP+90]. I find $l = 9$. Using this estimate for l, $\rho_d \approx 0.5995$ using the bisection method.

- I use a back-of-the-envelope calculation to calibrate Ψ, the relative variance of the demand shock to the fiscal shock. I calibrate Ψ so as to match the relative change in real GDP between Q1 and Q2 2020 vs the change in total debt outstanding during the same period. Using data from FRED (Federal Reserve Economic Data) I find that between Q1 and Q2 2020, total federal debt outstanding rose by $\approx 14\%$ and real GDP (chained 2012 US dollars) fell by $\approx 8.5\%$. Thus:

  $$\Psi \approx \frac{-8.5\%}{14\%} \approx -0.606$$

To facilitate comparisons, all other parameters are the same relative to the baseline calibration in Section 2.2.

3.3 (iii) Simulation: Optimal Monetary Policy

3.3.1 (a) Optimal Ramsey Policy

Under optimal Ramsey policy, policymakers set a path for the nominal interest rate $\{i_t\}_{t=0}^{\infty}$ so as to minimize the same loss function as in Section 2.3.1.

Figure 3 plots (in red) the impulse responses of the endogenous variables under optimal Ramsey policy to a correlated realization of the fiscal and demand shock. For reference, Figure 5 in the appendix compares the IRFs under optimal Ramsey policy between the one-shock/baseline case and the two-shock case for a fiscal shock of the same size. The optimal policy response is very similar but differs importantly in magnitude. In the two-shock model, the rise in the real value of the stock of debt on impact is approximately 3 times as large, thus it is intuitive that the policy response should be larger by a similar magnitude. Intuitively, the demand shock introduces an additional trade-off for the policymaker which makes it harder to engineer financial repression. The main takeaway from this exercise is that in the two-shock model, which more realistically exhibits the dynamic of the COVID-19 shock, engineering financial repression may be even harder than what is implied by the baseline model. This accentuates our concerns regarding the infeasibility of optimal Ramsey policy.

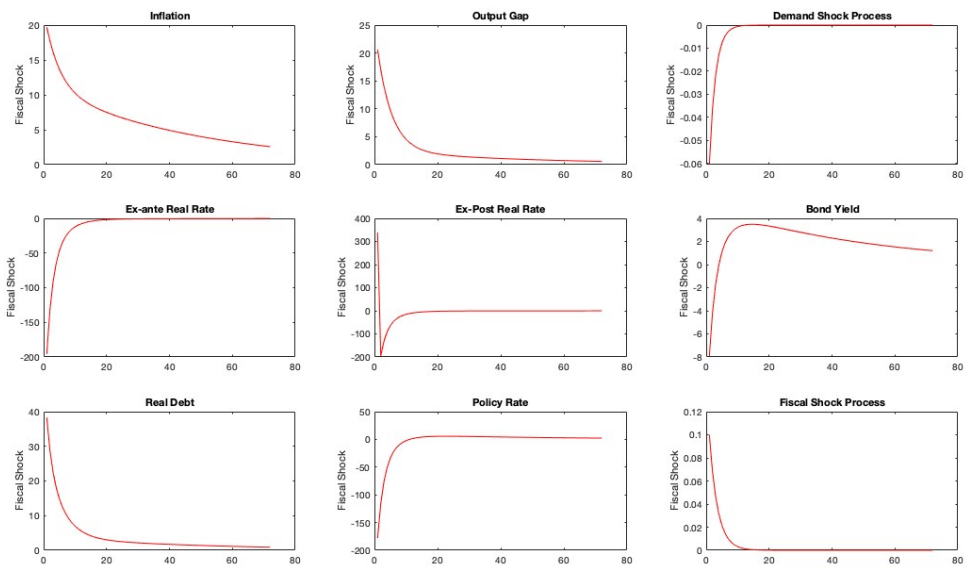

Figure 3: Optimal Ramsey Policy- Two-Schock Model

3.3.2 (b) Policy-Smoothed Taylor Rule

The policy-smoothed Taylor rule is the same as in Section 2.3.2, as well as its calibration, to facilitate comparisons.

Figure 4 plots the impulse responses of the endogenous variables under the policy-smoothing Taylor rule in response to a fiscal shock of the same magnitude as in the optimal Ramsey policy case. For reference, Figure 6 in the appendix compares the IRFs under the interest rate smoothing Taylor rule between the one-shock/baseline case and the two-shock case for a fiscal shock of the same size. The dynamics are similar to the policy smoothing case in the baseline model, in the sense that the strategy for absorbing the fiscal shock acts primarily through ex-post real rates rather than ex-ante rates as in the optimal Ramsey setting. However, in the two-shock model, there is the added complexity that the demand shock causes a fall in the output gap on impact. Thus policy rates can no longer be raised aggressively as this would amplify the negative response of output to the shock. Policy rates thus fall initially and are only subsequently brought to levels significantly above steady-state, as in the baseline model under interest rate smoothing. As a result, inflation rises more on impact and remains higher

throughout relative to its path in the model without the demand shock, ending the projection horizon $\approx 7\%$ above steady-state levels.

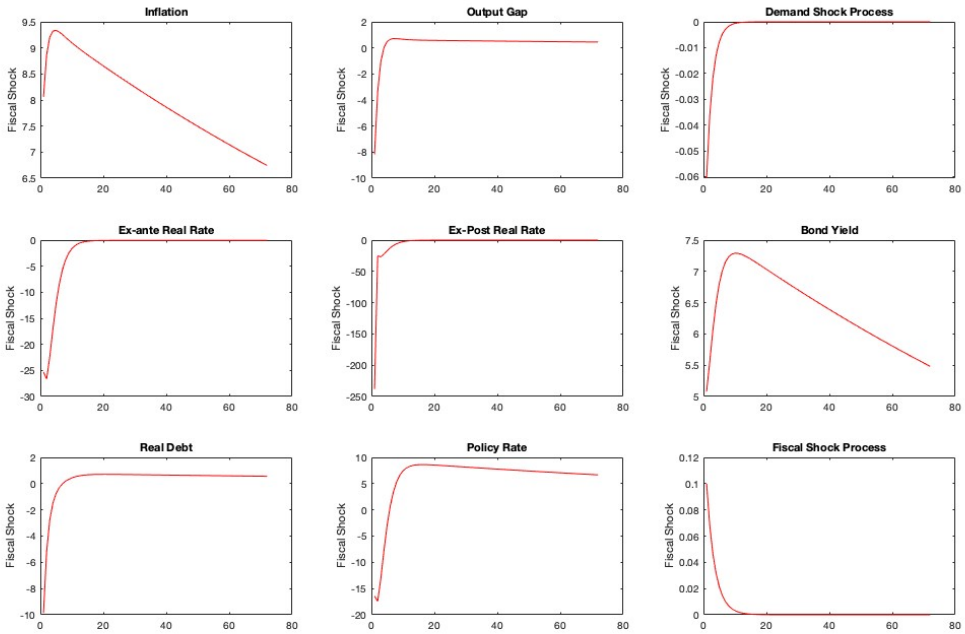

Figure 4: Policy smoothing Taylor Rule- Two-Shock Model

In contrast to Section 2.3.2, whereby introducing the policy smoothing relieves us from hitting the lower bound constraint in the baseline model, in the two-shock model this does not occur as a result of the added trade-off caused by the demand shock. Thus the policy-smoothing rule is not free of the lower bound criticism put forward to the optimal Ramsey policy response, although its implementation in practice is still more straightforward. This reinforces our view that the model may be missing some key features which would allow policymakers to absorb the shock in practice without violating the ELB.

3.4 (iv) Sensitivity analysis

The only parameter I subject to the sensitivity analysis that is not present in the baseline model is ρ_d. Figure 12 in the appendix plots the IRFs in response to the fiscal shock under optimal Ramsey policy in the two-shock model. I study 3 cases. The first, in blue, is the case with no persistence in the demand shock $\rho_d = 0$. The second, in red, is the baseline calibration of $\rho_d = 0.5995$. Lastly, in green, I set $\rho_d = 0.7743$, which corresponds to a value of $l = 18$, ie the persistence consistent with the demand shock taking double the time relative to the baseline calibration to (almost) fade away completely.

In all 3 cases, the optimal policy response on impact is pretty similar. However, as $\rho_d \to 1$, the policy rate becomes more sluggish in returning to steady-state values. This is because a larger ρ_d implies a larger decrease in the bond yield, which implies a larger initial rise in the real value of the debt stock through the ex-post return channel. This requires a more sluggish return of policy rates to steady-state levels after the shock.

4 Conclusion

In conclusion, this paper aims to extend the takeaways from the optimal policy in response to a fiscal shock along two dimensions.

The first dimension is, for a given 'optimal policy framework', ie optimal Ramsey policy or the interest rate smoothing Taylor rule, what are the main takeaways of optimal policy between the

baseline model vs the two-shock model. Under optimal Ramsey policy, I find that including the demand shock accentuates the fall in the policy rate needed to absorb the shock. Intuitively, the demand shock introduces an additional contemporaneous source of welfare loss for policymakers, the negative effect of the shock on the output gap, which induces the policymaker to respond to the shock more aggressively in order to contemporaneously reduce the real debt burden while stabilizing the real macroeconomic variables in its reaction function. In other words, the two-shock framework makes it much harder for policymakers to effectively engineer financial repression in practice. Under interest rate smoothing, the policy response is slightly different in the two models. Whilst in the baseline model, policy rates rise to absorb the shock through the ex-post return channel, with a limited role played by the ex-ante channel, this result is overturned in the two-shock model as the fall in the output gap caused by the demand shock calls for a cut in policy rates initially. Thus interest rate smoothing in the two-shock model employs some combination of the ex-post and ex-ante channels to absorb the shock.

The second dimension is, for a given model, what are the takeaways from the different policy frameworks, ie optimal Ramsey policy versus the interest rate smoothing Taylor rule. The differences between the two policies in the baseline model effectively boil down to the relative importance of the ex-ante return channel (financial repression) vs. the ex-post return channel. The former engineers financial repression, ie works through the ex-ante channel, while the latter works primarily through the ex-post channel. Cumulative welfare losses in the baseline model are only 2% higher in the interest rate smoothing case relative to optimal Ramsey policy, validating the findings from [Woo03]. In the two-shock model, optimal Ramsey policy still works through the ex-ante channel, while in the smoothing case, policy works through a combination of the two channels. Thus, while in the baseline model the responses under interest rate smoothing give us more realistic predictions, this is overturned in the two-shock model, highlighting how under the correlated realization it is much harder for policymakers to absorb the shock in practice.

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5 Appendix

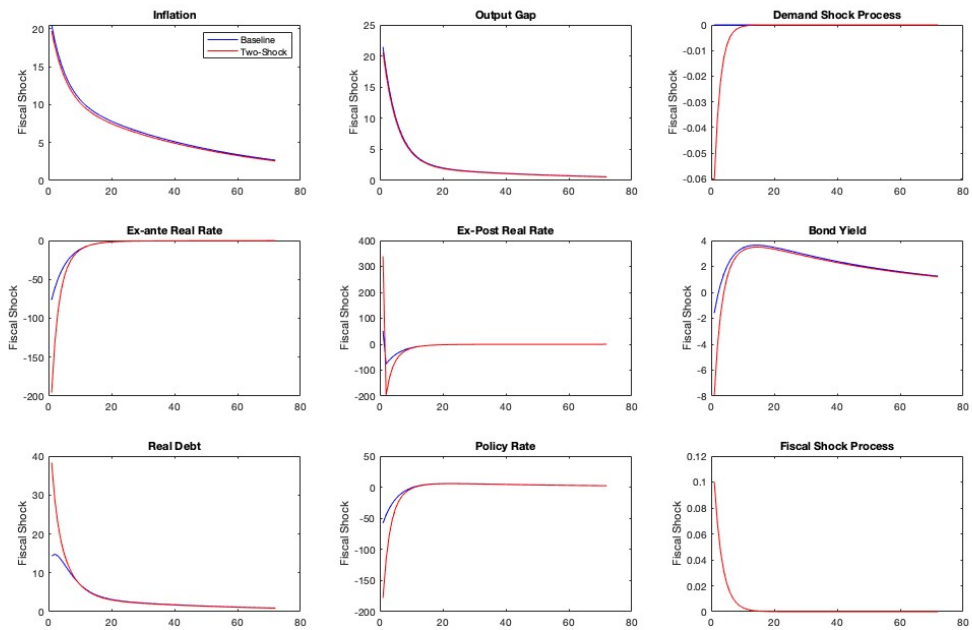

Figure 5: Optimal Ramsey Policy in Baseline vs Two-Shock model

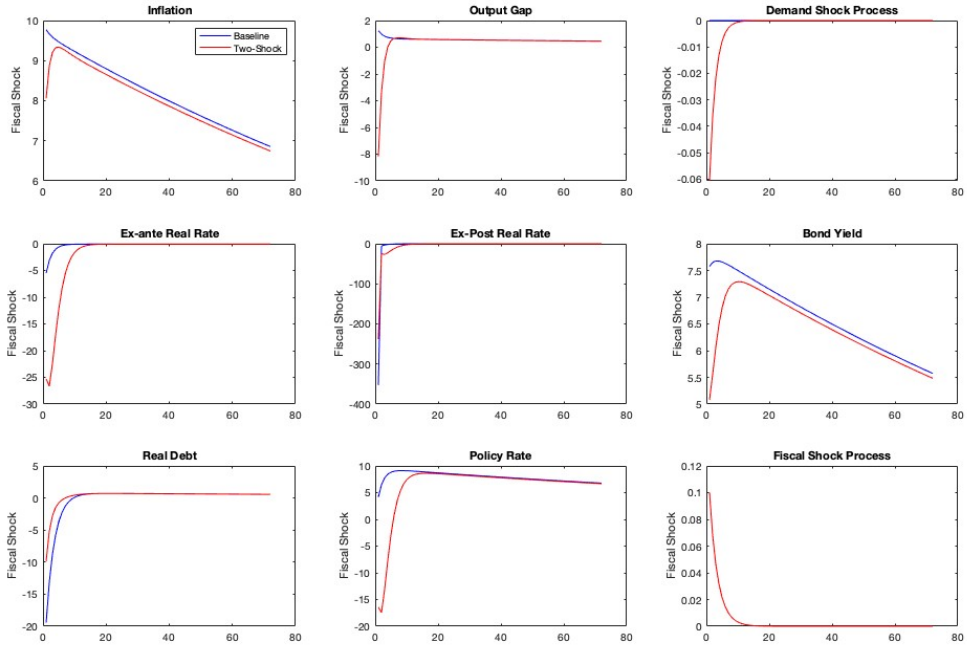

Figure 6: Interest rate smoothing Taylor Rule in Baseline vs Two-Shock model

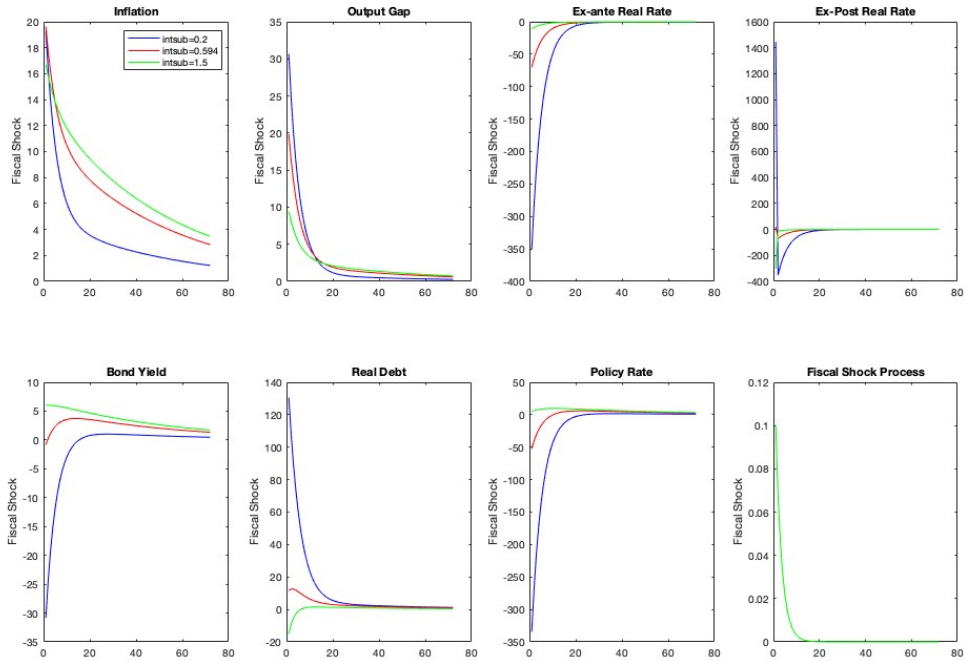

Figure 7: Optimal Ramsey Policy- Impact of σ in Baseline

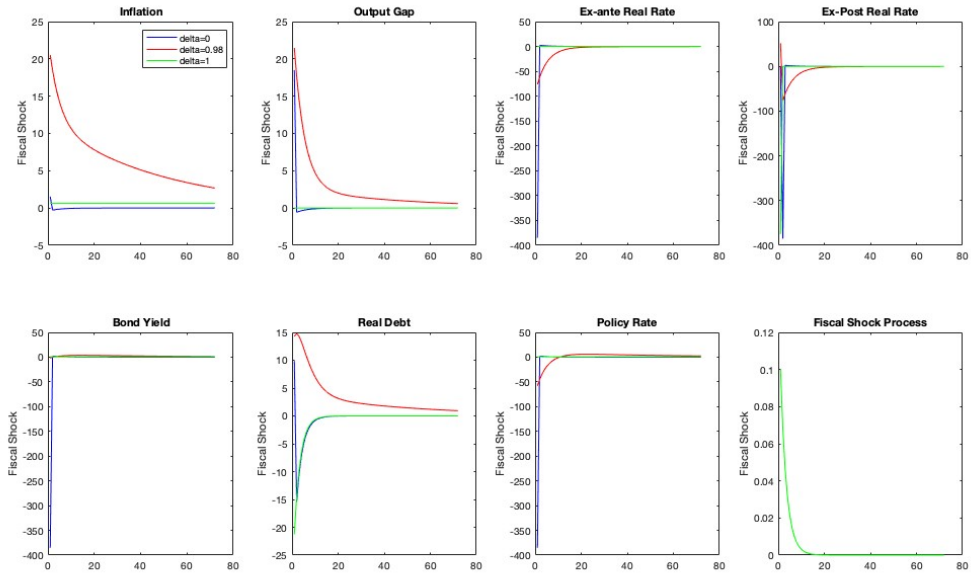

Figure 8: Optimal Ramsey Policy- Impact of δ in Baseline

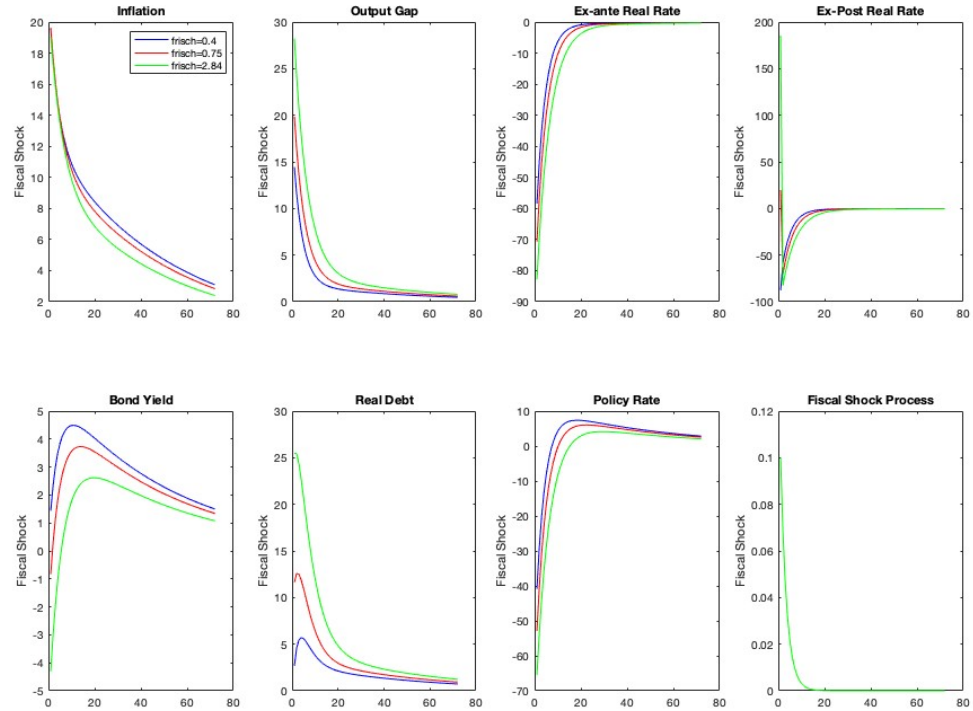

Figure 9: Optimal Ramsey Policy- Impact of ψ in Baseline

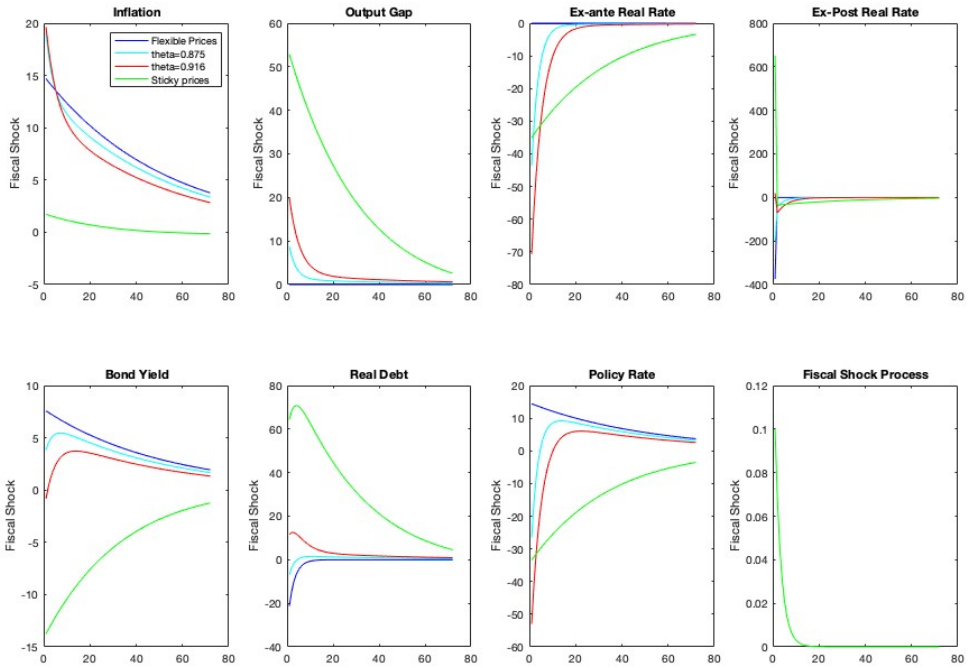

Figure 10: Optimal Ramsey Policy- Impact of θ in Baseline

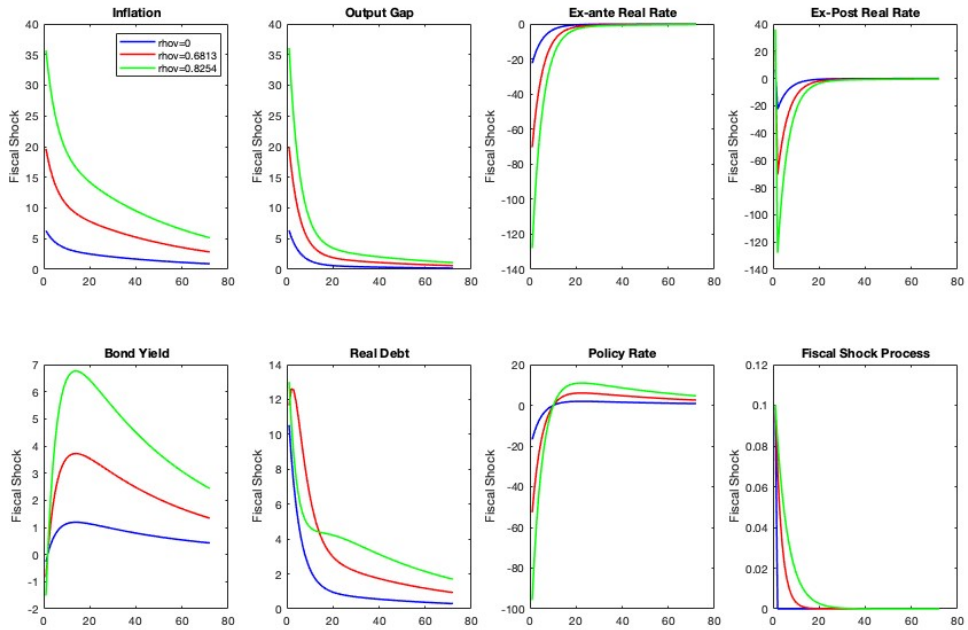

Figure 11: Optimal Ramsey Policy- Impact of ρ_v in Baseline

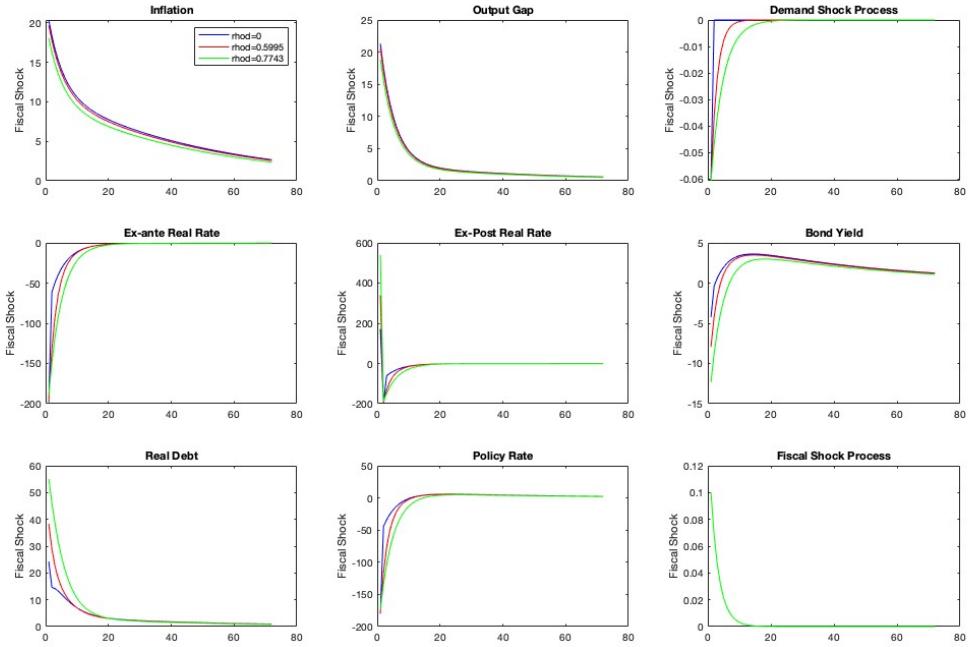

Figure 12: Optimal Ramsey Policy- Impact of ρ_d in Two-Shock model